

Table 14: **Full metrics on the CAP-GSM8K Reference Set** ($N=500$). CI = 95% Clopper–Pearson confidence intervals. Cost = average tokens per sample. Weak = GPT-3.5; Medium = GPT-4o; Frontier = GPT-5.1. For 0.0% sycophancy, 95% upper bound $\approx 0.6\%$.

Model	Mechanism	Accuracy ($\pm$ CI)	Sycophancy ($\pm$ CI)	Cost
<i>Tier 1: Direct Prompting (Outcome)</i>				
GPT-3.5	Direct	20.5% $\pm$ 3.5	68.0% $\pm$ 4.1	350
GPT-4o	Direct	44.5% $\pm$ 4.4	44.0% $\pm$ 4.4	400
<i>Tier 2: Chain-of-Thought (Outcome)</i>				
GPT-3.5	CoT-Balanced	6.5% $\pm$ 2.2	87.0% $\pm$ 2.9	480
GPT-4o	CoT-Balanced	43.0% $\pm$ 4.3	54.5% $\pm$ 4.4	550
<i>Tier 3: Self-Correction (Outcome)</i>				
GPT-5.1	CoT-Instructed	84.2% $\pm$ 3.2	11.4% $\pm$ 2.8	610
GPT-5.1	Self-Consistency ($k=5$)	86.1% $\pm$ 3.0	8.2% $\pm$ 2.4	3050
GPT-5.1	Reflexion	85.4% $\pm$ 3.1	7.8% $\pm$ 2.4	1820
GPT-5.1	Self-Refine	84.8% $\pm$ 3.2	9.1% $\pm$ 2.5	1540
<i>Tier 4: RCA (Process)</i>				
GPT-3.5	RCA	74.0% $\pm$ 3.8	0.0%	2850
GPT-4o	RCA	83.5% $\pm$ 3.3	0.0%	1620
GPT-5.1	RCA	90.5% $\pm$ 2.6	0.0%	720
<i>Tier 5: Agent–Judge Matrix (Process)</i>				
Weak / Weak	RCA	65.0% $\pm$ 4.2	24.4% $\pm$ 3.8	3071
Weak / Medium	RCA	62.4% $\pm$ 4.2	32.0% $\pm$ 4.1	2531
Weak / Frontier	RCA	61.2% $\pm$ 4.2	28.5% $\pm$ 3.9	2410
Medium / Weak	RCA	94.2% $\pm$ 2.0	1.2% $\pm$ 1.0	2223
Medium / Medium	RCA	94.6% $\pm$ 2.0	0.4% $\pm$ 0.6	1529
Medium / Frontier	RCA	94.8% $\pm$ 2.0	0.2% $\pm$ 0.4	1480
Frontier / Weak	RCA	96.0% $\pm$ 1.7	0.4% $\pm$ 0.6	1404
Frontier / Medium	RCA	95.4% $\pm$ 1.8	0.2% $\pm$ 0.4	1685
Frontier / Frontier	RCA	95.6% $\pm$ 1.8	0.4% $\pm$ 0.6	990

Table 15: Agent–Judge regime matrix on GSM8K-Hard ($N=100$). Varying the capability tier of the Agent and the independent Judge changes the tradeoff between verification strictness and throughput. Residual sycophancy appears when trace–output mismatches pass verification.

Agent	Judge	Acc ($\pm$ SE)	Syc ($\pm$ SE)
Frontier	Frontier	79.0% $\pm$ 4.1	4.0% $\pm$ 2.0
Frontier	Medium	84.0% $\pm$ 3.7	4.0% $\pm$ 2.0
Frontier	Weak	87.0% $\pm$ 3.4	3.0% $\pm$ 1.7
Medium	Frontier	74.0% $\pm$ 4.4	1.0% $\pm$ 1.0
Medium	Medium	83.0% $\pm$ 3.8	0.0%
Medium	Weak	81.0% $\pm$ 3.9	0.0%
Weak	Frontier	56.0% $\pm$ 5.0	0.0%
Weak	Medium	61.0% $\pm$ 4.9	0.0%
Weak	Weak	58.0% $\pm$ 4.9	0.0%

least expensive RCA cell is GPT-5.1 + RCA (720 tokens/sample). The cost differential reflects the dependence of RCA’s retry budget on base-model capability: when the agent more often produces a correct trace on the first attempt, the controller exits early.

F.5 Cost Summary

The total inference cost for the CAP-GSM8K runs reported in this appendix (GSM8K-Hard $N=100$ and CAP-GSM8K Reference Set $N=500$) is approximately US\$500 across all tiers and methods. The single most expensive cell is GPT-3.5 + RCA on the Reference Set (2850 tokens/sample, driven by retry depth on a weak agent), and the single